

100 Days of Linux - Day 044

Title: Practical Assignment - Compression & Archiving

Today's Goal

Apply everything learned about archiving and compression in one realistic backup and restore workflow. No new commands are introduced today.

Why This Matters

Developers, system administrators, and data engineers regularly back up projects, compress logs, archive datasets, and restore files. Today's assignment simulates these real-world tasks.

Commands Covered

tar, gzip, gunzip, zip, unzip

Scenario

You are responsible for backing up a small data engineering project before deployment. Create archives, compress files, verify backups, and restore them successfully.

Practical Assignment (5 Tasks)

1. Create a folder named `data_pipeline` containing `data/`, `logs/`, and `scripts/` subfolders. Add `employees.csv`, `sales.csv`, `application.log`, and `run.sh` with sample content.
2. Create a TAR archive named `data_pipeline.tar`. List its contents without extracting it.
3. Create a compressed archive named `data_pipeline.tar.gz`. Extract it into a folder named `restore_tar` and verify the restored files.
4. Create a ZIP archive named `data_pipeline.zip`. List its contents, then extract it into a folder named `restore_zip`.
5. Compress `application.log` using `gzip` while keeping the original file. Verify the `.gz` file exists, then display your current working directory.

Hints

Complete one task at a time. Verify every archive before extracting it using `tar -tvf` or `unzip -l`.

Mini Challenge

Create a backup folder containing two projects. Archive one as `.tar.gz` and the other as `.zip`. Extract both into separate restore folders and compare the restored structures.

Common Mistakes

Trying to compress directories directly with `gzip`. Use `tar` with the `-z` option for folders.

Did You Know?

Many automated backup systems generate .tar.gz archives every night to protect production data.

Pro Tip

Always verify an archive before deleting the original files. A quick inspection can prevent accidental data loss.

Answer Key (Instructor Only)

Q1: `mkdir -p data_pipeline/{data,logs,scripts}` and create sample files

Q2: `tar -cvf data_pipeline.tar data_pipeline/ && tar -tvf data_pipeline.tar`

Q3: `tar -czvf data_pipeline.tar.gz data_pipeline/ && mkdir restore_tar && tar -xzvf data_pipeline.tar.gz -C restore_tar/`

Q4: `zip -r data_pipeline.zip data_pipeline/ && unzip -l data_pipeline.zip && mkdir restore_zip && unzip data_pipeline.zip -d restore_zip/`

Q5: `gzip -k data_pipeline/logs/application.log && pwd && ls data_pipeline/logs/*.gz`

Homework

Repeat today's workflow from memory using a different project folder until you can complete every step confidently.

Next Day Preview

Tomorrow you'll begin the Processes module by learning how Linux runs programs and how to inspect running processes using `ps`.